CSE 2109 – Electronic Engineering

50 Most Important MCQs – BJT, Transistor, FET, JFET & MOSFET

Selected from the uploaded class materials | Answer is given immediately below each question

⚠️ **Exam Focus:** These are high-priority questions based on the uploaded notes. They are selected for exam preparation, but no question can be guaranteed to appear exactly in the exam.

BJT / Transistor Basics

1. What does BJT stand for?

A) Bipolar Junction Transistor
B) Binary Junction Transistor
C) Base Junction Technology
D) Bipolar Joint Terminal
Answer: A

2. How many terminals does a BJT have?

A) 2
B) 3
C) 4
D) 5
Answer: B

3. Which three terminals belong to a BJT?

A) Gate, Source, Drain
B) Anode, Cathode, Gate
C) Emitter, Base, Collector
D) Input, Output, Ground
Answer: C

4. A transistor consists essentially of how many pn junctions?

A) One
B) Two
C) Three
D) Four
Answer: B

5. A transistor can be regarded as a combination of:

A) Two diodes connected back to back
B) Two resistors in series
C) Two capacitors
D) One diode and one capacitor
Answer: A

6. Which region of a BJT is very thin and lightly doped?

A) Emitter
B) Base
C) Collector
D) Substrate

Answer: B

7. Which BJT region is heavily doped to inject a large number of charge carriers?

A) Base
B) Collector
C) Emitter
D) Junction
Answer: C

8. Which BJT region is moderately doped and physically larger to dissipate heat?

A) Base
B) Collector
C) Emitter
D) Gate
Answer: B

9. What are the main charge carriers in an NPN transistor?

A) Holes
B) Electrons
C) Ions
D) Photons
Answer: B

10. What are the main charge carriers in a PNP transistor?

A) Electrons
B) Holes
C) Neutrons
D) Ions
Answer: B

11. Which equation represents the transistor current relationship?

A) $I_E = I_C - I_B$
B) $I_E = I_B + I_C$
C) $I_C = I_E + I_B$
D) $I_B = I_E + I_C$
Answer: B

12. In normal transistor action, approximately what percentage of emitter-injected carriers cross into the collector according to the notes?

A) Less than 5%
B) About 25%
C) More than 95%
D) 50%
Answer: C

13. In the active region of a BJT, the emitter-base junction is ___ and collector-base junction is ___.

A) reverse biased, reverse biased
B) forward biased, reverse biased
C) reverse biased, forward biased
D) forward biased, forward biased
Answer: B

14. In the saturation region of a BJT, both junctions are:

A) Reverse biased
B) Forward biased
C) Open
D) Unbiased
Answer: B

15. In the cut-off region of a BJT, both junctions are:

A) Forward biased
B) Reverse biased
C) Shorted
D) Saturated
Answer: B

16. In which BJT region does the transistor normally act as an amplifier?

A) Cut-off
B) Saturation
C) Active
D) Breakdown
Answer: C

17. In which two BJT regions does the transistor act as a switch?

A) Active and breakdown
B) Cut-off and saturation
C) Active and saturation
D) Cut-off and active
Answer: B

18. At cut-off, the collector-emitter voltage is approximately:

A) 0 V
B) VCC
C) VBE
D) βVCC
Answer: B

19. What is the current amplification factor α in common-base configuration?

A) IC/IB
B) IE/IB
C) IC/IE
D) IB/IC
Answer: C

20. The typical range of α given in the notes is:

A) 0.01–0.1
B) 0.2–0.5
C) 0.9–0.99
D) 20–500
Answer: C

21. What is the current amplification factor β in common-emitter configuration?

A) IC/IB

B) IE/IC
C) IB/IE
D) IC/IE
Answer: A

22. The typical range of β given in the notes is:
A) 0.9–0.99
B) 1–5
C) 20–500
D) 750–1000
Answer: C

23. Which relation between α and β is correct?
A) $\beta = \alpha/(1-\alpha)$
B) $\beta = 1-\alpha$
C) $\beta = \alpha+1$
D) $\beta = 1/\alpha$
Answer: A

24. If α = 0.96, what is β?
A) 12
B) 24
C) 48
D) 96
Answer: B

25. Which BJT configuration is used in about 90–95% of transistor applications according to the notes?
A) Common base
B) Common emitter
C) Common collector
D) Common gate
Answer: B

26. What is the main reason the common-emitter configuration is widely used?
A) Very low current gain
B) High current, voltage and power gain
C) No amplification
D) Only low input resistance
Answer: B

27. The input resistance of a common-base circuit is of the order of:
A) A few ohms
B) A few hundred kΩ
C) Several MΩ
D) 750 kΩ
Answer: A

28. The output resistance of a common-base circuit is of the order of:
A) A few ohms
B) Several tens of kΩ
C) A few Ω

D) 20 Ω
Answer: B

29. The input resistance of a common-emitter circuit is of the order of:
A) A few hundred ohms
B) Several MΩ
C) 750 kΩ
D) 25 Ω
Answer: A

30. The common-collector configuration has approximately:
A) Low input and high output resistance
B) Very high input and very low output resistance
C) Zero input resistance
D) Very high output resistance only
Answer: B

FET / JFET

31. What does FET stand for?
A) Field-Effect Transistor
B) Forward Electron Transistor
C) Field Energy Terminal
D) Frequency Effect Transistor
Answer: A

32. A FET is primarily a:
A) Current-controlled device
B) Voltage-controlled device
C) Temperature-controlled device
D) Power-controlled device
Answer: B

33. Which three terminals are found in a FET?
A) Emitter, Base, Collector
B) Anode, Cathode, Gate
C) Source, Drain, Gate
D) Input, Output, Base
Answer: C

34. A key characteristic of a FET is:
A) Very low input impedance
B) High input impedance
C) High base current
D) Low reliability
Answer: B

35. The JFET controls its channel using a:
A) Forward-biased p-n junction
B) Reverse-biased p-n junction
C) Metal contact

D) Transformer
Answer: B

36. According to the notes, JFET operates in:
A) Enhancement mode only
B) Depletion mode only
C) Both only at zero bias
D) Saturation mode only
Answer: B

37. In an n-channel JFET, the main conduction path is made of:
A) p-type material
B) n-type material
C) oxide
D) metal
Answer: B

38. In an n-channel JFET with VGS = 0 and VDS > 0, which current equality is given?
A) ID = 0
B) ID = IS
C) ID = IG
D) IS = 0
Answer: B

39. For the JFET condition VGS = 0 V, the gate current IG is:
A) Maximum
B) Equal to ID
C) 0 A
D) Negative ID
Answer: C

40. At low VDS, a JFET operates in the:
A) Ohmic region
B) Breakdown region
C) Cut-off region
D) Inverted region
Answer: A

41. At pinch-off, the depletion regions:
A) Disappear completely
B) Nearly meet near the drain
C) Become zero near the source
D) Become forward biased
Answer: B

42. In the JFET saturation region, increasing VDS further generally:
A) Increases ID significantly
B) Does not increase ID
C) Makes IG large
D) Makes the channel wider
Answer: B

43. IDSS is defined as:

A) Minimum gate current
B) Maximum drain current when VGS = 0 V
C) Drain current at VDS = 0
D) Gate-source saturation voltage
Answer: B

44. Which equation is Shockley's equation for a JFET?

A) ID = IDSS(1 − VGS/VP)^2
B) ID = βIB
C) ID = VGS/R
D) ID = IDSS + VGS
Answer: A

45. For a JFET, at VGS = VP, Shockley's equation gives:

A) ID = IDSS
B) ID = IDSS/2
C) ID = 0
D) ID = 1 A
Answer: C

46. For a JFET, at VGS = VP/2, ID is:

A) IDSS
B) IDSS/2
C) IDSS/4
D) 0
Answer: C

MOSFET

47. What does MOSFET stand for?

A) Metal-Oxide-Semiconductor Field-Effect Transistor
B) Metal Output Silicon Field Electronic Transistor
C) Main Oxide Semiconductor Frequency Transistor
D) Metal-Oriented Signal FET
Answer: A

48. Which MOSFET mode is normally OFF at VGS = 0?

A) Depletion mode
B) Enhancement mode
C) Saturation mode
D) Ohmic mode
Answer: B

49. Which MOSFET mode is normally ON at VGS = 0?

A) Enhancement mode
B) Depletion mode
C) Cut-off mode
D) Triode mode
Answer: B

50. Which are the two MOSFET channel types?

A) N-channel and P-channel
B) E-channel and B-channel
C) Source-channel and drain-channel
D) Alpha-channel and beta-channel
Answer: A

51. The gate of a MOSFET is insulated, resulting in:

A) Low input impedance
B) High input impedance
C) High base current
D) Large gate current
Answer: B

52. NMOS stands for:

A) Negative Metal Oxide System
B) N-channel MOSFET
C) New Mode Operating Switch
D) N-type Mechanical Oxide Semiconductor
Answer: B

53. For an enhancement-type NMOS, conduction begins when VGS is:

A) Below VTN
B) Equal to 0 only
C) Greater than or equal to VTN
D) Always negative
Answer: C

54. For an enhancement NMOS in the saturation region, the condition is:

A) $VDS \geq VGS - VTN$
B) $VDS = 0$
C) $VGS < VTN$
D) $VDS < 0$
Answer: A

55. For an enhancement NMOS in saturation, the drain current is given by:

A) $ID = Kn(VGS - VTN)$
B) ID = 1/2 Kn(VGS − VTN)^2
C) $ID = VDS/R$
D) $ID = \beta IB$
Answer: B

56. The MOSFET overdrive voltage is:

A) $VOV = VGS + VTN$
B) $VOV = VGS - VTN$
C) $VOV = VDS - VGS$
D) $VOV = VTN - VGS$
Answer: B

57. For an enhancement NMOS in the triode/linear region, the condition is:

A) $VGS \geq VTN$ and $0 \leq VDS \leq VGS - VTN$

B) VGS < VTN
C) VDS > VGS
D) VGS = 0 only
Answer: A

58. Which device is described in the notes as a foundation of modern digital integrated circuits?
A) JFET
B) MOSFET
C) Vacuum tube
D) Diode
Answer: B

59. In the MOSFET example, if VTN = 1 V, VGS = 3 V and VDS = 4.5 V, the transistor is in:
A) Cut-off
B) Triode only
C) Saturation
D) Breakdown
Answer: C

60. In the MOSFET example, if VTN = 1 V, VGS = 3 V and VDS = 1 V, the drain current is given in the notes as:
A) 0.2 mA
B) 0.6 mA
C) 1.0 mA
D) 2.0 mA
Answer: B

61. For the worked NMOS example with VTN = 0.4 V and Kn = 1.40 mA/V^2, at VGS = 0.8 V, ID is approximately:
A) 0.024 mA
B) 0.224 mA
C) 1.40 mA
D) 2.02 mA
Answer: B

62. For the same worked NMOS example, at VGS = 1.6 V, ID is approximately:
A) 0.224 mA
B) 0.80 mA
C) 2.02 mA
D) 4.48 mA
Answer: C

Source Basis

BJT lecture: introduction, construction, working regions, α/β, configurations, characteristics, Q-point and transistor states.

Transistor lecture: transistor structure, carrier action, common-base/common-emitter/common-collector configurations, α/β/γ, load-line and switching states.

FET lecture: FET characteristics, JFET operation, pinch-off, saturation, IDSS and Shockley equation.

MOSFET lecture: enhancement/depletion modes, NMOS/PMOS, regions of operation, threshold/overdrive voltage and drain-current equations/examples.